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Examination	Institute	Year	CPI/%
Graduation	Indian Institute of Technology Bombay	2025 – present	9.68
Intermediate	Narayana e-Techno School	2025	96.8%
Matriculation	Wisdom High International School	2023	99.2%

Pursuing a **Minor in Statistics**

Research interests: randomised algorithms, probabilistic method, derandomisation, complexity classes

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 7** in **JEE Advanced** among 180,000+ candidates 2025
- Achieved **All India Rank 102** in **JEE Main** among 1,400,000+ candidates 2025
- Shortlisted for the **Aditya Birla Scholarship** interview 2026
- Awarded the **Gold Medal** in the **Homi Bhabha Balvaidnyanik Competition** 2019-20
- Secured **State Rank 5** in the **Maharashtra Scholarship Exam** among 200,000+ students 2018-19

OLYMPIADS & COMPETITIONS

- Among the **top 600** of 10,000+ participants to reach the in-person regionals of **IICPC QuantFest**, and secured **Rank 227** of 600 at the regionals 2026
- Among the **top 300** of 40,000+ candidates in the National Standard Examination in Physics (NSEP), selected for the **Indian National Physics Olympiad (INPhO)**, HBCSE 2024-25
- Cleared the National Standard Examination in Astronomy (NSEA) among 40,000+ candidates to qualify for the **Indian National Astronomy Olympiad (INAO)**, HBCSE 2024-25
- Cleared the National Standard Examination in Chemistry (NSEC) among 40,000+ candidates to qualify for the **Indian National Chemistry Olympiad (INChO)**, HBCSE 2024-25
- Secured **1st position** in **Physics Bazinga** and **Maths Bazinga**, Maths & Physics Club 2025
- 3rd** in the quizzing round of **The Graviton Crucible**, IIT Bombay Edition 2026

PROFESSIONAL EXPERIENCE

First-Year Trading and Technology Program (FTTP) | *Jane Street* (2026)
Hong Kong

- Shortlisted for **Jane Street's** selective program for first-year STEM undergraduates
- Attended classes on Jane Street's approach to **trading and technology** and competed in a **team-based mock trading simulation**
- Developed **fair-value estimates** for a complex **live market simulation** with market inefficiencies

KEY PROJECT | RESEARCH-ORIENTED

Randomised Algorithms: A 137-Page Monograph | *Summer of Science* (2026)
Maths and Physics Club, IIT Bombay GitHub: [Sahil-M-Deo/Randomised-Algorithms](#)

- Authored a self-contained **137-page, 13-chapter book** in L^AT_EX as an extension of the project report, with **6 appendices** on probability, concentration inequalities, limit theorems, finite fields and linear algebra, and **original end-of-chapter problem sets**
- Covered **RP, ZPP & BPP**, analysis of QuickSort and QuickSelect, **universal & k -wise independent hashing**, **Freivalds' algorithm**, skip lists, treaps, Bloom filters, cuckoo hashing, **Miller–Rabin**, **Pollard's rho**, streaming (**Flajolet–Martin**, **HyperLogLog**), zero-knowledge proofs and **derandomisation**

- Using the **probabilistic method**, designed Las Vegas algorithms for dominating set and max-cut; proved a success probability of at least $\frac{1}{m+1}$ for finding a cut with $\geq m/2$ edges, giving $\Theta(nm)$ expected time
- **Conjectured** from large-scale simulation (exhaustive enumeration for small n , Monte Carlo sampling for large n) that the **5-cycle is the worst-case graph** for the randomised **max-cut** algorithm, with success probability $\frac{5}{16}$; this would tighten the bound to $\Theta(n+m)$ expected time
- Proved via a **rank argument** on ± 1 orthogonal columns that generating n pairwise-independent bits needs $\geq \log_2(n+1)$ truly random bits, tight for the **XOR construction**; derandomised max-cut via pairwise independence and the **method of conditional expectations**
- Used **Monte Carlo sampling** (10,000 trials per n) to validate an **approximate-independence** assumption in a probabilistic construction, and to show the **union bound** is near-tight in practice

OTHER PROJECTS

Deep Chess | Seasons of Code

Web and Coding Club, IIT Bombay

- Built a C++ chess engine rated **2200**, benchmarked in matches against **Stockfish**, with **alpha-beta search**, iterative deepening, **quiescence search** and a **Zobrist-hashed transposition table**, using MVV-LVA and killer-move ordering
- Trained an **NNUE evaluation network** in **PyTorch** with incremental accumulator updates for fast inference, benchmarked against the classical evaluation in **cutechess** tournaments

Reinforcement Learning for Algorithmic Trading | Winter in Data Science

Analytics Club, IIT Bombay

- Compared **greedy**, ϵ -**greedy** and **UCB** strategies on multi-armed bandits and modelled environments as **MDPs**
- Implemented a **Q-Learning** stock trading agent on historical price data and evaluated it against a **buy-and-hold** baseline

Mini Game Hub | Course Project: Software Systems Lab (CS108)

- Built a two-player hub in **Bash** and **Python/pygame** with Tic-Tac-Toe, Othello and Connect Four, time controls and persistent leaderboards
- Implemented user authentication with **SHA-256** password hashing and account lockout after repeated failed attempts

Game Theory | Learners' Space

Maths and Physics Club, IIT Bombay

- Studied Nash equilibria, mixed strategies and the **minimax principle**, applied in the monograph to motivate randomisation against adversarial inputs

TECHNICAL SKILLS & COURSES

Languages	C++, Python, Java, JavaScript, Bash, Verilog, Assembly, L ^A T _E X
Tools	Git, Make, Pygame, NumPy, Z3 Solver (CNF-SAT)
Competitive Programming	Rated 1654 on Codeforces
Courses	Computer Programming and Utilisation (CS101), Discrete Structures (CS105), Software Systems Lab (CS108)
Ongoing Courses	Data Structures & Algorithms (CS213) [†] , Digital Logic Design & Computer Architecture (CS230) [†] , Logic for Computer Science (CS228), Data Analysis & Interpretation (CS215)
	[†] with corresponding lab (CS293, CS232)